ABCfold NPF Conformation Pipeline

Six structure-prediction backends, launched together, clustered by TM-helix alignment

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Outline

1. Background
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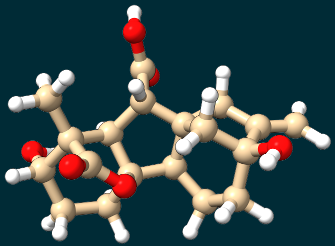
Gibberellins

- Key phytohormone in plants; isolated in 1931 from *Gibberella fujikuroi* (fungus), later found in plants (1950)
- Promotes growth and development: stem elongation, seed germination, flowering
- GA-deficiency mutants (key oxidase disrupted): smaller but more resilient, **higher grain yield**

⇒ GA-deficiency-linked yield gains underpinned the **Green Revolution** (1970s)



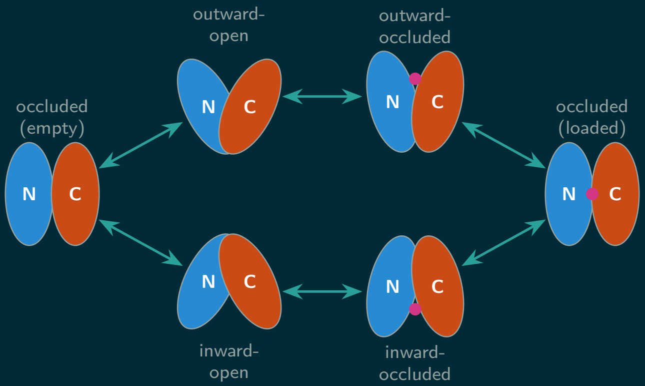
Gibberellin A1 — $C_{19}H_{24}O_6$, 348.4 g/mol (PubChem CID 5280379)

Nitrate and Peptide Transporters (NPF)

- Part of the **Major Facilitator Superfamily (MFS)** of membrane transporters
- Transport nitrate (NO_3^-), dipeptides, and gibberellins
- Tissue- and substrate-specific

MFS transport cycle: six conformations

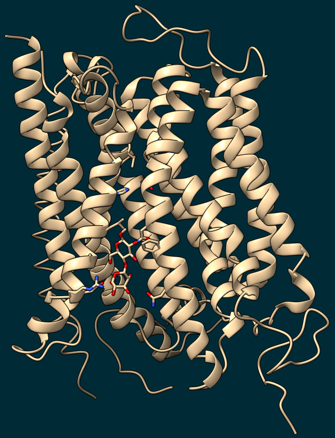
Alternating-access mechanism: transport requires passing through a fully **occluded** intermediate in either direction — the **biological reason** the pipeline targets six conformations per protein, not six independent opinions on one.



Redrawn after Drew *et al.*, *Chem. Rev.* 2021, 121, 5289–5335 (Fig. 8a), CC BY-NC-ND 4.0 — DOI:10.1021/acs.chemrev.0c00983

What's experimentally known

- Only **2 NPFs** have experimentally resolved structures: NPF6.3 (4OH3, 5A2N, 5A2O), NPF2.10 (9UI1, 9UI6, 9UIF, 9UIT)
- Only **importers** are well characterized in the literature (technical constraints)
- **Exporters** remain unidentified



NPF6.3 — PDB 4OH3, X-ray, 3.25 Å
Sun *et al.*, *Nature* 2014

Research project goals

1. **Ab initio modelling of 3D structures**
    1.1 Pipeline to generate the six conformations *(this project)*
    1.2 Network interaction analysis (ligand and intra-chain)
2. **Functional annotation of NPFs**
    2.1 Identify key features of importers vs. non-importers
    2.2 Eventually, find exporters

The question

- Arabidopsis **NPF** (Nitrate/Peptide Transporter) family

  Which conformations can each NPF transporter adopt, apo and ligand-bound?
- Structure prediction as a proxy for conformational sampling

Why not just one model?

- Sibling repo `AF3_NPF_pipeline` already sampled **30 AF3 seeds/protein**
- Overlaying those ensembles with Boltz-2 (Gibberellin importers, 8 proteins) showed: AF3 alone **does not** recover every conformation a second, architecturally-different model finds

$\Rightarrow$ One model isn't enough — run **six together** instead of resampling one

The six backends

- AlphaFold3
- Boltz-2
- Chai-1
- OpenFold3
- Protenix
- RosettaFold3

All launched **together** per protein via ABCfold's `abcfold -abcopr` against the *same* resolved fold input.

Three stages

Pre-processing (local, needs internet)
sequences → fold inputs → MMseqs2 MSA

↓ rsync fold_inputs/

Processing (IFB cluster, no internet)
ABCfold: 6 backends launched together

↓ rsync results/abcfold/

Post-processing (local)
DeepTMHMM → TM-helix alignment → clustering

Stage 1–3: from sequence to resolved fold input

1. **Sequences** — UniProt → per-protein FASTA
2. **Fold input** — `fold_input.json` per protein × apo/holo (AF3-dialect JSON, ABCfold's native input: sequence + every replica seed)
3. **MMseqs2 default run** — MSA + top-hit templates from the ColabFold webserver, no manual curation, no pocket restraint; embedded into `fold_input.resolved.json`

Runs locally, driven by Snakemake (`worflows/preprocessing/Snakefile`); resumable — existing resolved inputs are never re-fetched.

Stage 4: ABCfold on the IFB cluster

- One `abcfold -abcopr...` call per protein × form
- Launches all 6 backends against the same `fold_input.resolved.json`
- AlphaFold3 backend runs via IFB's `alphafold` module (Singularity wrapper, auto-discovered) — no separate image needed
- Boltz/Chai-1/OpenFold3/Protenix/RosettaFold3 built once via `--prime` on a login node (needs internet)

Gibberellin importers run first

- 8 GA1-ligand proteins (`NPF2.1`, `NPF2.5`, `NPF2.10`, `NPF2.12`, `NPF2.13`, `NPF3.1`, `NPF4.1`, `NPF5.6`) — the group already studied with AF3 + Boltz-2
- Listed in `priority_gibberellin.txt`, put first in the SLURM array manifest
- Fastest path to validating the six-backend approach against a known result

Stage 5–6: topology and alignment

1. **DeepTMHMM** — TM-helix topology (BioLib), once
2. **TM-helix Kabsch alignment** — per protein × form, pooled across every backend and seed

Output: results/tm_alignment/<protein>/{aligned_ca_tm.npy, meta.csv} — meta.csv now carries a **model** column (alphafold3/boltz/chai1/openfold3/protenix/rosettafold3).

Stage 7: clustering & visualisation

- Notebook-based (JupyterLab, `envs/notebook.yaml`)
- Same schema as `AF3_NPF_pipeline`'s clustering notebooks + one new axis: **color/facet by backend**
- *That's the whole point of this pipeline*: which backend(s) actually reach which conformational cluster?
- Status:

Where we are

-
-
-

Next steps

Questions?